COMPATIBLE BOUNDED-ABOVE RESOLUTIONS WITH SPLIT TRANSITION MAPS FOR AN ARBITRARY COMPLEX

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Editorial note. P is only closed under finite direct sums and epimorphism-generating; its members need not be projective. The split-injection homotopy adjustment and final contractible correction remain counted work.

Editorial scope. One original target with its local construction proved in full. Classical category primitives are explicitly allowed.

Allowed standard prerequisites (editorial).

- Bounded-above resolutions by an epimorphism-generating class. A bounded-above complex has a termwise surjective quasi-isomorphism from a bounded-above complex in P if every object is a quotient of an object in P.
- Homotopy-category triangles. The homotopy category of an abelian category is triangulated; any distinguished triangle is isomorphic to one arising from a termwise split exact sequence. Standard cone formulas and long exact cohomology are allowed.

Source reference. Original proposition 014S: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 3319–3326. Standard structure of the homotopy category. Stable Stacks tag displayed.

Source reference. Original lemma 05T7: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 5301–5316. Explicit permitted single-complex bounded-above resolution theorem; compatibility and split tower repair are not granted. Stable Stacks tag displayed.

Source reference. Original lemma 0G6C: focused-seeds/ACL/sources/stacks-snapshot-20261011/derived.tex, source lines 3458–3466. Standard split-cone realization of a homotopy triangle. Stable Stacks tag displayed.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{P} \subset \mathrm{Ob}(\mathcal{A})$ be a subset. Assume $\mathcal{P}$ contains 0, is closed under (finite) direct sums, and every object of $\mathcal{A}$ is a quotient of an element of $\mathcal{P}$. Let $K^\bullet$ be a complex. There exists a commutative diagram*

$$\begin{array}{ccccc} P_1^\bullet & \longrightarrow & P_2^\bullet & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \\ \tau_{\leq 1}K^\bullet & \longrightarrow & \tau_{\leq 2}K^\bullet & \longrightarrow & \ldots \end{array}$$

in the category of complexes such that

(1) *the vertical arrows are quasi-isomorphisms and termwise surjective,*
(2) *$P_n^\bullet$ is a bounded above complex with terms in $\mathcal{P}$,*
(3) *the arrows $P_n^\bullet \to P_{n+1}^\bullet$ are termwise split injections and each cokernel P_{n+1}^i/P_n^i is an element of $\mathcal{P}$.*

Date: stacks-project:derived.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. We are going to use that the homotopy category $K(\mathcal{A})$ is a triangulated category, see Proposition 014S. By Lemma 05T7 we can find a termwise surjective map of complexes $P_1^\bullet \to \tau_{\leq 1}K^\bullet$ which is a quasi-isomorphism such that the terms of $P_1^\bullet$ are in $\mathcal{P}$. By induction it suffices, given $P_1^\bullet, \ldots, P_n^\bullet$ to construct $P_{n+1}^\bullet$ and the maps $P_n^\bullet \to P_{n+1}^\bullet$ and $P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$.

Choose a distinguished triangle $P_n^\bullet \to \tau_{\leq n+1}K^\bullet \to C^\bullet \to P_n^\bullet[1]$ in $K(\mathcal{A})$. Applying Lemma 05T7 we choose a map of complexes $Q^\bullet \to C^\bullet$ which is a quasi-isomorphism such that the terms of $Q^\bullet$ are in $\mathcal{P}$. By the axioms of triangulated categories we may fit the composition $Q^\bullet \to C^\bullet \to P_n^\bullet[1]$ into a distinguished triangle $P_n^\bullet \to P_{n+1}^\bullet \to Q^\bullet \to P_n^\bullet[1]$ in $K(\mathcal{A})$. By Lemma 0G6C we may and do assume $0 \to P_n^\bullet \to P_{n+1}^\bullet \to Q^\bullet \to 0$ is a termwise split short exact sequence. This implies that the terms of $P_{n+1}^\bullet$ are in $\mathcal{P}$ and that $P_n^\bullet \to P_{n+1}^\bullet$ is a termwise split injection whose cokernels are in $\mathcal{P}$. By the axioms of triangulated categories we obtain a map of distinguished triangles

$$\begin{array}{ccccccc} P_n^\bullet & \longrightarrow & P_{n+1}^\bullet & \longrightarrow & Q^\bullet & \longrightarrow & P_n^\bullet[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ P_n^\bullet & \longrightarrow & \tau_{\leq n+1}K^\bullet & \longrightarrow & C^\bullet & \longrightarrow & P_n^\bullet[1] \end{array}$$

in the triangulated category $K(\mathcal{A})$. Choose an actual morphism of complexes $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$. The left square of the diagram above commutes up to homotopy, but as $P_n^\bullet \to P_{n+1}^\bullet$ is a termwise split injection we can lift the homotopy and modify our choice of f to make it commute. Finally, f is a quasi-isomorphism, because both $P_n^\bullet \to P_n^\bullet$ and $Q^\bullet \to C^\bullet$ are.

At this point we have all the properties we want, except we don't know that the map $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$ is termwise surjective. Since we have the commutative diagram

$$\begin{array}{ccc} P_n^\bullet & \longrightarrow & P_{n+1}^\bullet \\ \downarrow & & \downarrow \\ \tau_{\leq n}K^\bullet & \longrightarrow & \tau_{\leq n+1}K^\bullet \end{array}$$

of complexes, by induction hypothesis we see that f is surjective on terms in all degrees except possibly n and $n+1$. Choose an object $P \in \mathcal{P}$ and a surjection $q : P \to K^n$. Consider the map

$$g : P^\bullet = (\ldots \to 0 \to P \xrightarrow{1} P \to 0 \to \ldots) \longrightarrow \tau_{\leq n+1}K^\bullet$$

with first copy of P in degree n and maps given by q in degree n and $d_K \circ q$ in degree $n+1$. This is a surjection in degree n and the cokernel in degree $n+1$ is $H^{n+1}(\tau_{\leq n+1}K^\bullet)$; to see this recall that $\tau_{\leq n+1}K^\bullet$ has $\mathrm{Ker}(d_K^{n+1})$ in degree $n+1$. However, since f is a quasi-isomorphism we know that $H^{n+1}(f)$ is surjective. Hence after replacing $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$ by $f \oplus g : P_{n+1}^\bullet \oplus P^\bullet \to \tau_{\leq n+1}K^\bullet$ we win. $\square$